UNSTEADY FLOW MODELS

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Windkessel Models

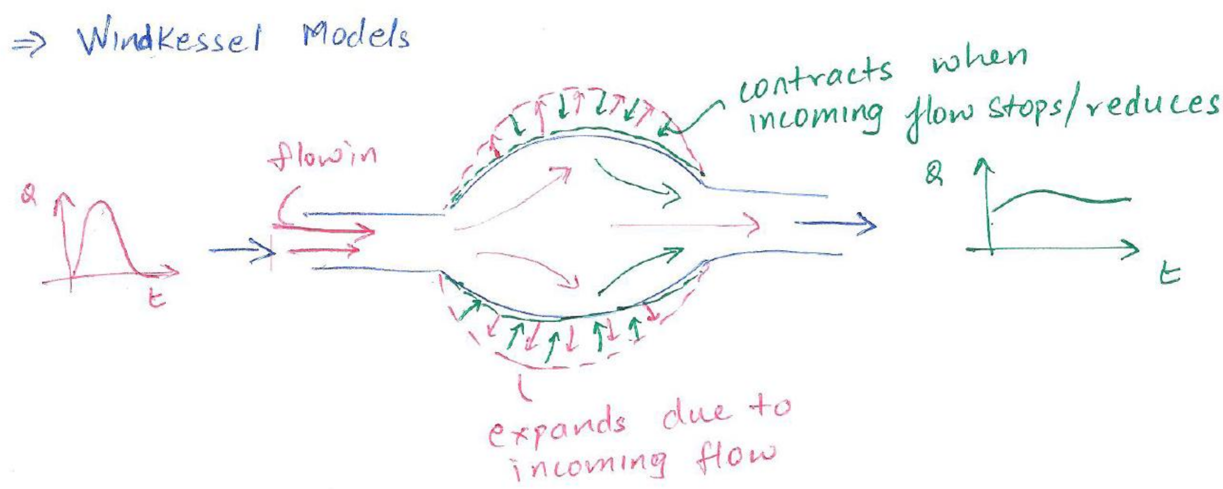

Figure 1: Representation of expansion and contraction of compliant vessel

- The vessel:
    - expands due to incoming flow
    - contracts when incoming flow stops/reduces

Vascular Resistance

$$\text{Resistance} = \frac{\Delta P}{Q}$$

- Smaller radius → higher pressure drop:

$$R_2 < R_1 \Rightarrow \Delta P_2 > \Delta P$$

Units:

$$[mmHg - s/ml]$$

Distensibility (Capacitance)

$$D = \frac{\Delta V}{\Delta P} \text{ OR } C = \frac{\Delta V}{\Delta P}$$

Units:

$$[ml/mmHg]$$

Electrical Analogy of Circulatory System

- Ohm's law:

$$V = IR$$

Analogy:

- $I = Q$(flow)
- $V = \Delta P$(pressure difference)
- R =vascular resistance

Flow Model Components

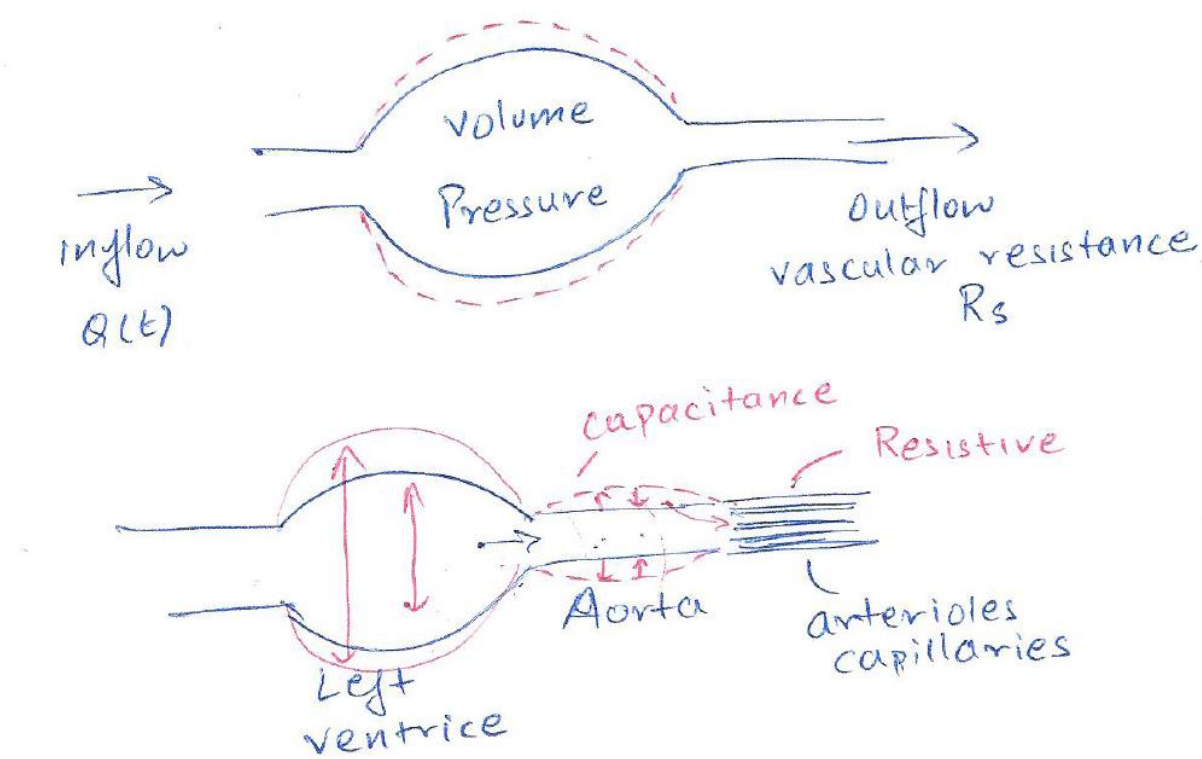

Figure 2: Windkessel model of arterial system

- Inflow: $Q_{in}(t)$
- Outflow
- Vascular resistance: R_s
- Capacitance: C

Flow balance:

$$Q_{in} - Q_{out} = \frac{d(\text{Vol})}{dt}$$

Since:

$$Q_{out} = \frac{P}{\text{Res}}$$
$$Q_{in} - \frac{P}{\text{Res}} = \frac{d(\text{Vol})}{dt}$$

Capacitance Relation

$$\frac{d(\text{Vol})}{dt} = \frac{d(\text{Vol})}{dP}\frac{dP}{dt}$$
$$\frac{d(\text{Vol})}{dt} = C\frac{dP}{dt}$$

Governing Equation

$$Q_{in} - \frac{P}{\text{Res}} = C\frac{dP}{dt}$$

Rearranged:

$$C\frac{dP}{dt} + \frac{P}{\text{Res}} = Q_{in}(t)$$
$$\frac{dP}{dt} + \frac{P}{\text{Res} \cdot C} = \frac{Q_{in}(t)}{C}$$

- This is a **linear differential equation for pressure**

Square Wave Input Flow

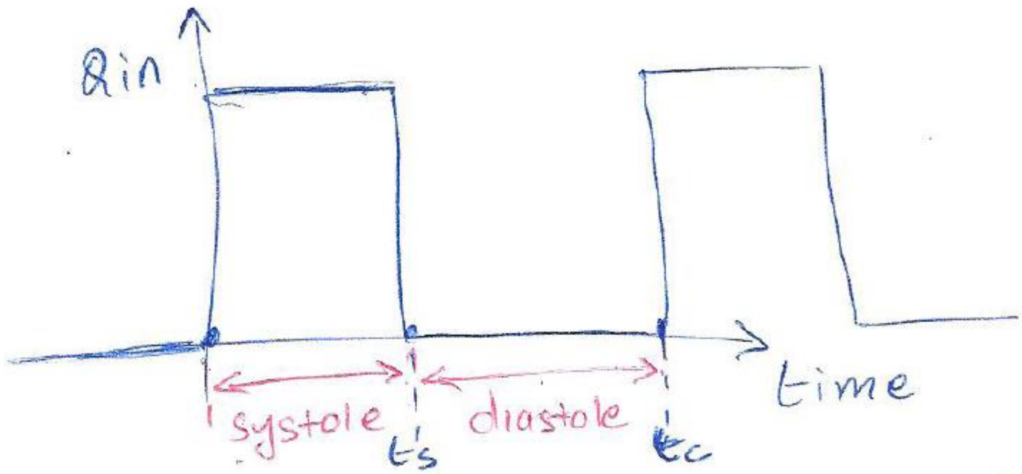

Figure 3: Square wave input

Assume:

$$Q_{in}(t)$$

is a square wave.

- Systole:

$$0 < t < t_s, Q_{in} = Q_0$$

- Diastole:

$$t_s < t < t_c, Q_{in} = 0$$

Governing Equations

Systole ($\mathbf{0 < t < t_s}$)

$$\frac{dP}{dt} + \frac{P}{\text{Res} \cdot C} = \frac{Q_0}{C}$$

Diastole ($\mathbf{t_s < t < t_c}$)

$$\frac{dP}{dt} + \frac{P}{\text{Res} \cdot C} = 0$$

Time Constant

$$\tau = \text{Res} \cdot C$$

Solution Using Integrating Factor

Integrating factor:

$$IF = e^{t/(\text{Res} \cdot C)}$$

Systole Solution

Initial condition:

$$P(0) = P_0$$

(where P_0= end diastolic pressure)

$$P(t) = P_0 + Q_0 \cdot \text{Res}\left(1 - e^{-t/(\text{Res} \cdot C)}\right)$$

Diastole Solution

$$\frac{dP}{dt} = -\frac{P}{\text{Res} \cdot C}$$
$$\frac{dP}{P} = -\frac{dt}{\text{Res} \cdot C}$$
$$P = C_1 e^{-t/(\text{Res} \cdot C)}$$

Initial condition:

$$P(t_s) = ESP$$

(End systolic pressure)

$$P(t) = ESP \cdot e^{-t/(\text{Res} \cdot C)}$$

Final Solution

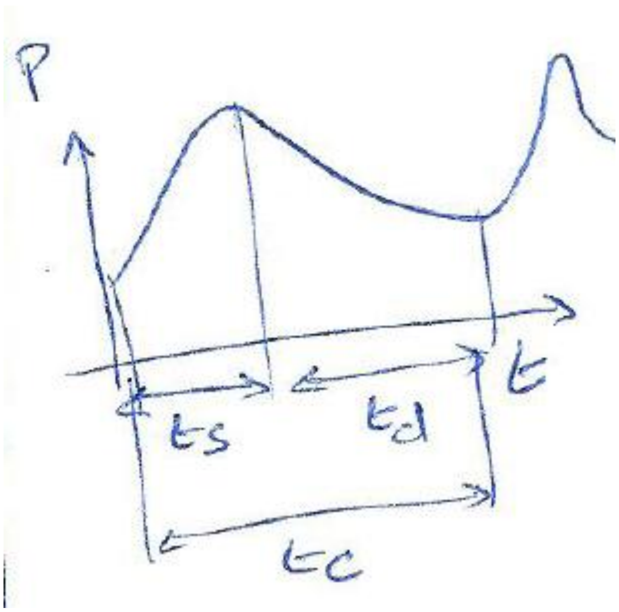

Figure 4: Time-varying pressure output of Windkessel model

For systole ($0 < t < t_s$)

$$P(t) = P_0 + Q_0 \cdot \mathrm{Res}\left(1 - e^{-t/(\mathrm{Res}\cdot C)}\right)$$

For diastole ($t_s < t < t_c$)

$$P(t) = ESP \cdot e^{-t/(\mathrm{Res}\cdot C)}$$

Electrical Circuit Analogy

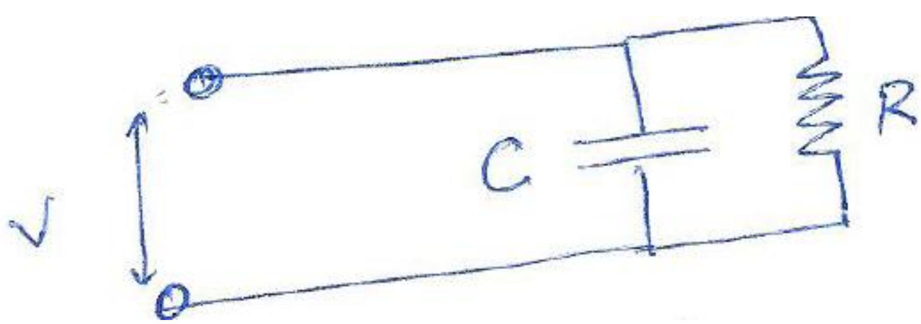

Figure 5: Simple R-C circuit

Circuit equation:

$$I(t) = \frac{V(t)}{R} + C\frac{dV(t)}{dt}$$

Using analogy:

- $I = Q$
- $V = P$
- $R = \text{Res}$

$$Q(t) = \frac{P(t)}{\text{Res}} + C\frac{dP(t)}{dt}$$

Rearranged:

$$\frac{dP}{dt} + \frac{P(t)}{\text{Res} \cdot C} = \frac{Q(t)}{C}$$

2–Element Windkessel Model

- Represents the human circulatory system using:
  - Resistance (R)
  - Capacitance (C)

Flow and Pressure Behavior

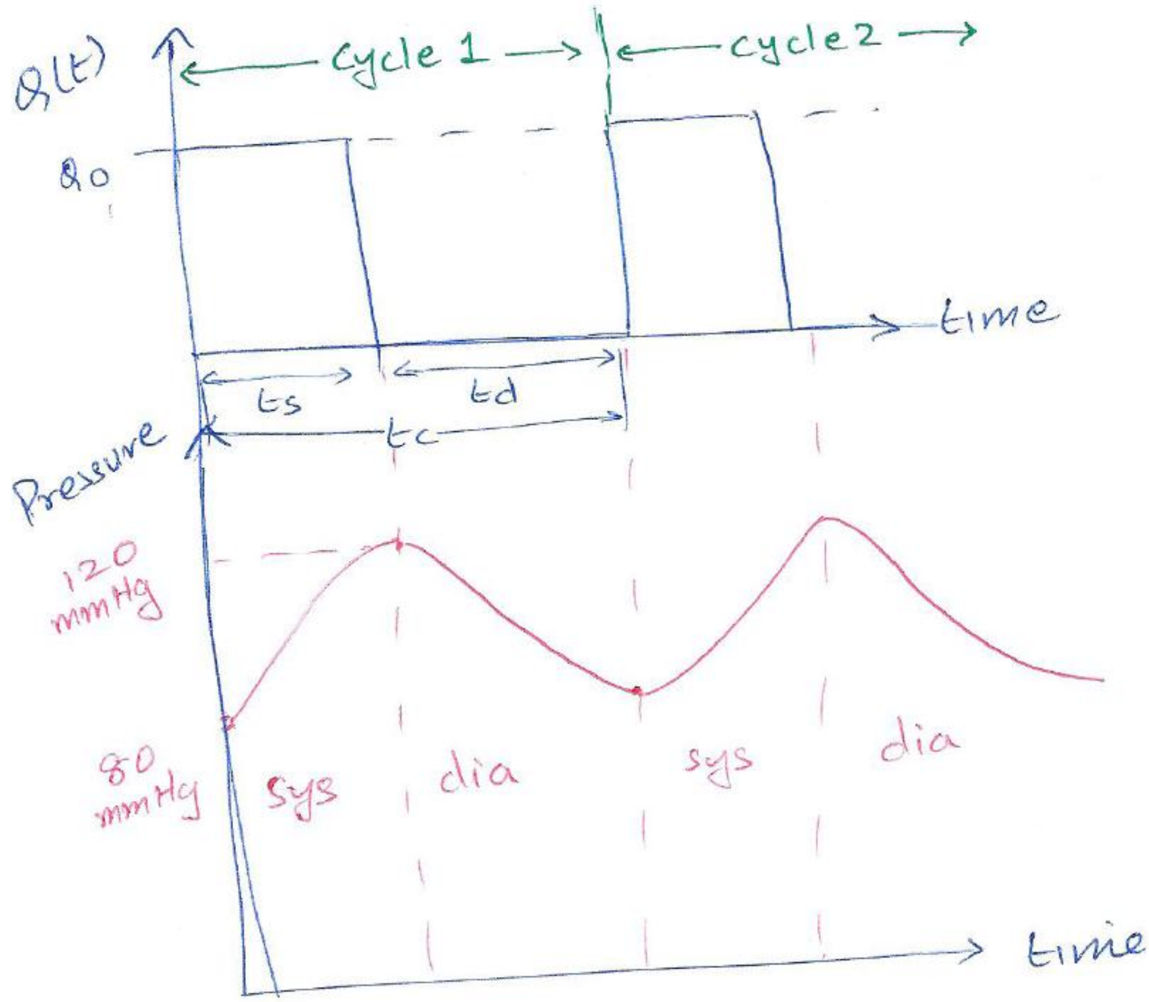

Figure 6: Representation of time-varying pressure output for square wave input to Windkessel model

- Flow:
  - periodic square-wave input
  - cycles: systole → diastole → repeat
- Pressure:
  - rises during systole
  - decays exponentially during diastole
